Technological University of the Philippines Visayas
COLLEGE OF ENGINEERING
Talisay City, Negros Occidental

Bi-directional SRAM Testbench

A Project presented to the

Computer Engineering Department

of Technological University of the Philippines Visayas

In Partial Fulfillment

of the Requirements for the Subject

HDL

Tappa, Nestor Luis Jr. M.

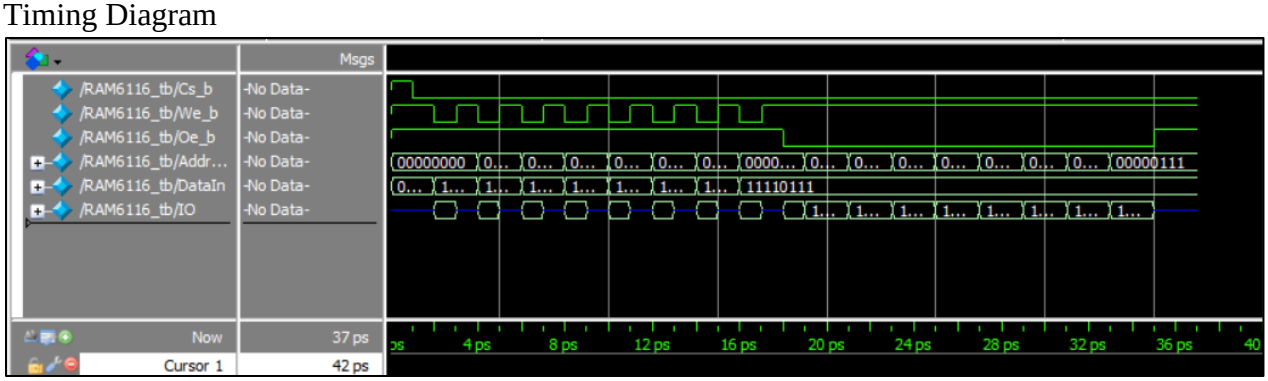
MODELSIM SIMULATION

Verilog Code

```
module RAM6116 (Cs_b, We_b, Oe_b, Address, IO);

        input Cs_b, We_b, Oe_b;
        input [7:0] Address;
        inout [7:0] IO;

        reg [7:0] RAM [0:255];

        assign IO = (Cs_b == 1'b1 || We_b == 1'b0 || Oe_b == 1'b1) ? 8'bzzzzzzzz : RAM[Address];

        always @(We_b, Cs_b) begin

                @ (negedge We_b)
                if (Cs_b == 1'b0) begin
                        RAM[Address] <= IO;
                end
        end

endmodule

module RAM6116_tb;

        reg Cs_b, We_b, Oe_b;
        reg [7:0] Address;
        reg [7:0] DataIn;
        wire [7:0] IO;

        RAM6116 dut (Cs_b, We_b, Oe_b, Address, IO);

        assign IO = (!We_b && !Cs_b) ? DataIn : 8'bZZZZZZZZ;

        initial begin
                Cs_b = 1;
                We_b = 1;
                Oe_b = 1;
                Address = 8'b00000000;
                DataIn = 8'b00000000;
        end
```

```
        initial begin
                #1 Cs_b = 0;

                #1 Address = 8'b00000000; DataIn = 8'b11110000; We_b = 0;
                #1 We_b = 1;
                #1 Address = 8'b00000001; DataIn = 8'b11110001; We_b = 0;
                #1 We_b = 1;
                #1 Address = 8'b00000010; DataIn = 8'b11110010; We_b = 0;
                #1 We_b = 1;
                #1 Address = 8'b00000011; DataIn = 8'b11110011; We_b = 0;
                #1 We_b = 1;
                #1 Address = 8'b00000100; DataIn = 8'b11110100; We_b = 0;
                #1 We_b = 1;
                #1 Address = 8'b00000101; DataIn = 8'b11110101; We_b = 0;
                #1 We_b = 1;
                #1 Address = 8'b00000110; DataIn = 8'b11110110; We_b = 0;
                #1 We_b = 1;
                #1 Address = 8'b00000111; DataIn = 8'b11110111; We_b = 0;
                #1 We_b = 1;

                #1 Oe_b = 0;
                #1 Address = 8'b00000000;
                #1 $display("The data at address %b is %b", Address, IO);
                #1 Address = 8'b00000001;
                #1 $display("The data at address %b is %b", Address, IO);
                #1 Address = 8'b00000010;
                #1 $display("The data at address %b is %b", Address, IO);
                #1 Address = 8'b00000011;
                #1 $display("The data at address %b is %b", Address, IO);
                #1 Address = 8'b00000100;
                #1 $display("The data at address %b is %b", Address, IO);
                #1 Address = 8'b00000101;
                #1 $display("The data at address %b is %b", Address, IO);
                #1 Address = 8'b00000110;
                #1 $display("The data at address %b is %b", Address, IO);
                #1 Address = 8'b00000111;
                #1 $display("The data at address %b is %b", Address, IO);

                #1 Oe_b = 1;
                #1 $display("The data at address %b is %b", Address, IO);

                #1 $finish;
        end
endmodule
```

Timing Diagram



Transcript

```
#
# The data at address 00000000 is 11110000
# The data at address 00000001 is 11110001
# The data at address 00000010 is 11110010
# The data at address 00000011 is 11110011
# The data at address 00000100 is 11110100
# The data at address 00000101 is 11110101
# The data at address 00000110 is 11110110
# The data at address 00000111 is 11110111
# The data at address 00000111 is zzzzzzzz
# ** Note: $finish    : C:/intelFPGA_lite/20.1/SRAM/Bidirectional.v(81)
#    Time: 37 ps  Iteration: 0  Instance: /RAM6116_tb
# 1
# Break in Module RAM6116_tb at C:/intelFPGA_lite/20.1/SRAM/Bidirectional.v line 81
```